A balance lab for three eras of humanoid locomotion control: capture-point feedback, step MPC and a PPO policy on the same pushed pendulum

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September 6, 2026

Abstract

The Science Robotics survey *Evolution of Humanoid Locomotion Control* (Gu, Shi, Shi et al., 2026) organises sixty years of bipedal control into three eras and argues that they share three principles: physics-based modelling, constrained decision making and adaptation to uncertainty. This note builds the smallest experiment in which those principles can be pushed with a finger: a planar linear inverted pendulum (LIP) walker with foot placement, an ankle (ZMP) input, impulsive pushes, sensor noise and delay, and a true centre-of-mass height that may differ from the controller's model. Four controllers close the loop, one per paradigm: capture-point stepping with an ankle strategy at a fixed period (classical real-time feedback), the same law with the pendulum frequency estimated online (adaptive control), a three-step model predictive controller with step-length constraints and variable step timing (predictive control), and a 64×64 tanh policy trained by proximal policy optimisation on 512 parallel walkers with domain randomisation (learning-based control), plus the same policy trained on the nominal body only as the sim-to-real ablation (each the better of two seeds, chosen on the nominal body). Under a single push the fixed-period classical law survives 64.10 % of 1.00 m/s shoves and 23.40 % of 1.60 m/s ones; the predictive controller 100.00 % and 92.20 %; the randomised policy 100.00 % and 100.00 %. With the true centre of mass at 0.60 m instead of the modelled 0.80 m, under random pushes, the nominal-only policy falls in 6.25 % of episodes against 0.00 % for the randomised one: a well-trained policy with a velocity history in its observation adapts implicitly, and the difference between two training seeds of the nominal policy (41.00 % against 6.25 % at 0.60 m) was larger than the difference between the two training distributions. The lab is a mechanism demonstration on a reduced-order model, not a humanoid benchmark; it is the interactive core of the page `locomotion.html` and is not part of the survey.

Keywords: linear inverted pendulum, capture point, divergent component of motion, model predictive control, proximal policy optimisation, domain randomisation, humanoid locomotion.

1 Overview

The survey's Fig. 3 sorts controllers by when their optimisation happens (hardly, online, offline) and on what (reduced-order model, full model, simulator, data). The lab reproduces that axis on one model: the classical law computes nothing at run time beyond a formula; the predictive controller solves a small constrained problem every tick; the learned policy did all its optimisation offline in simulation and executes a network. The same disturbances, the same sensors and the same body are applied to all. Code: `experiments/lipwalk/` (`model.py`, `controllers.py`, `sim.py`), `train_rl.py`, `run_study.py`, `make_figures.py`, `make_videos.py`; browser port `assets/js/locomotion_lab.js` with the exported weights.

2 Method

2.1 The walker

Between steps the horizontal CoM position x relative to the stance foot follows the LIP (survey ref. 45)

$$\ddot{x} = \omega^2(x-a), \qquad \omega = \sqrt{g/z_0}, \qquad |a| \le a_{\max} = 0.05\,\mathrm{m}, \tag{1}$$

where a is the ZMP offset the ankle produces. With a held over a tick $\Delta t = 10.00\,\mathrm{ms}$ the map is exact: $x^+ = a + (x-a)\cosh\omega\Delta t + \dot{x}\,\sinh(\omega\Delta t)/\omega$, $\dot{x}^+ = (x-a)\,\omega\sinh\omega\Delta t + \dot{x}\cosh\omega\Delta t$. A step moves the stance foot by u ($|u| \le L_{\max} = 0.65\,\mathrm{m}$): $x \leftarrow x - u$, $\dot{x}$ unchanged, i.e. instantaneous swing and no impact loss (a point-foot LIP assumption). Steps are permitted after $T_{\min} = 0.24\,\mathrm{s}$ and forced at $T_{\max} = 0.60\,\mathrm{s}$; the nominal period is $T = 0.40\,\mathrm{s}$. Pushes are velocity impulses. The controller observes x exactly (a kinematic quantity), and $\dot{x}$ delayed by d ticks with Gaussian noise of standard deviation σ, plus the last four velocity samples. A fall is $|x| > 0.75\,\mathrm{m}$ or $|\dot{x}| > 3.00\,\mathrm{m/s}$. The nominal body has $z_0 = 0.80\,\mathrm{m}$ ($\omega = 3.50\,\mathrm{s}^{-1}$); the true z_0 is an experiment parameter.

The periodic gait. With the divergent component of motion $\xi = x + \dot{x}/\omega$ (the capture point), the end-of-step DCM of the periodic gait with stride $L = v_d T$ is $\xi_T = b\,e^{\omega T}$ with $b = L/(e^{\omega T}-1)$ the DCM right after the step. All controllers are started on this orbit.

2.2 Controllers

Classical (real-time feedback). Ankle strategy $a = k\,(\xi - b\,e^{\omega t})$ with $k = 0.8$ towards the nominal DCM trajectory; at the fixed step time T, $u = \xi_T - b$ (step to the capture point minus the periodic offset). Uses the nominal ω; myopic; the step is clipped to $L_{\max}$ if it cannot be taken.

Adaptive. The same law with ω^2 estimated by recursive least squares on $\dot{x}_{k+1} - \dot{x}_k = \omega^2(x_k - a_k)\Delta t$ (forgetting factor 0.98, updates suspended across steps).

Predictive (step MPC). Every tick, for each candidate step time $T_c \in \{0.26, 0.31, 0.36, 0.40, 0.46, 0.54\}$ s (a time already passed means: step now), propagate ξ to T_c and solve

$$\min_{u_1..u_3} \sum_{j=1}^{3}(\xi_j - b\,e^{\omega T})^2 + \rho\sum_k (u_k - L)^2, \quad \xi_j = e^{j\omega T}\xi_{T_c} - \sum_{k<j} e^{(j-k)\omega T}u_k, \quad |u_k| \le L_{\max}, \tag{2}$$

by two rounds of clip-and-resolve on the 3×3 normal equations ($\rho = 0.15$), add $0.3\,(T_c - T)^2/T^2$, and apply the first step of the cheapest candidate. The ankle law is the classical one.

Learned. Observation $(x, \dot{x}_{\mathrm{meas}}, t/T, v_d, u_{\mathrm{last}}, a/a_{\max}, \dot{x}_{k-3..k})$ (10 values); actions (a, u, T_c) through tanh squashing to their ranges. Reward per tick $0.01\,[1 - 1.5(\dot{x}-v_d)^2 - 0.3(a/a_{\max})^2]$, minus $0.02\,|u - v_d T|$ at a step, minus 4 at a fall (episode end). PPO (Schulman et al. 2017), 512 walkers, 48-tick rollouts, GAE $\lambda = 0.95$, $\gamma = 0.99$, clip 0.2, 4 epochs, minibatch 8192, Adam 3×10^{-4}, 500 iterations (≈ 12 M samples), actor and critic 64×64 tanh, running observation normalisation. *Domain randomisation* (policy `dr`): $z_0 \sim U[0.55, 1.05]$ m, $v_d \sim U[-0.3, 1.0]$ m/s, $\sigma \sim U[0, 0.2]$, $d \in \{1,2,3\}$ ticks, push size $\sim U[0.4, 1.5]$ m/s at 0.6 pushes/s. The `nodr` policy sees the nominal body, $\sigma = 0.05$, $d = 1$, pushes of 1 m/s, the same range of v_d.

3 Experiments

Protocols (`run_study.py`, 64 walkers per point, seeded): a single push of magnitude m at $t = 2$ s at a random step phase (survival within 6 s, time to settle within 0.12 m/s of v_d for 0.50 s); random pushes (0.6/s, up to 1.00 m/s) for 10 s while the true z_0 varies from 0.5 to 1.1 m with the model at 0.8 m; the same while σ varies from 0 to 0.4 m/s with 10 or 30 ms delay; velocity tracking across $v_d \in [-0.3, 1.0]$ m/s with gentle pushes; and one traced episode per controller under the shared push schedule $(+0.8, -1.0, +1.2, -0.6, +1.4)$ m/s at 1.5, 3.5, 5.5, 7.5, 9.5 s.

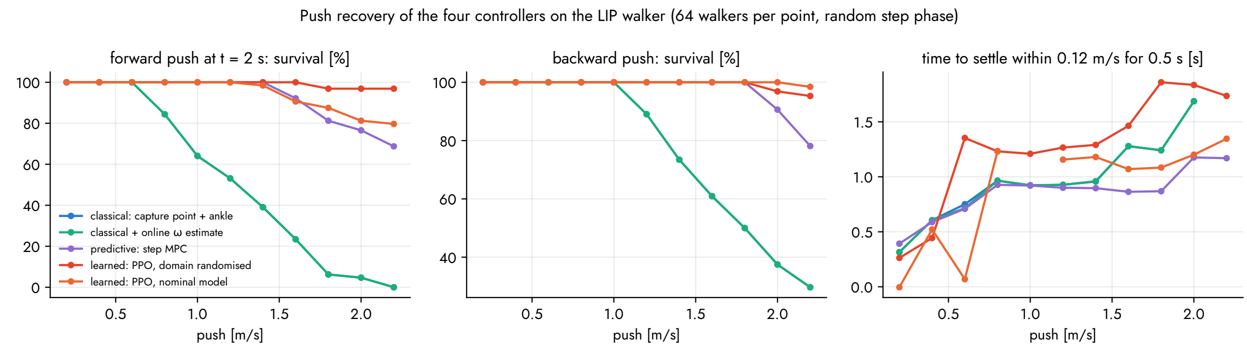

Figure 1: Push recovery. The fixed-period classical law survives half its forward pushes up to 1.20 m/s, the predictive controller up to 2.20 m/s and the randomised policy up to 2.20 m/s. Adaptation of ω does not change the classical curve: the fixed timing and the clipped step are the binding limits, not the model.

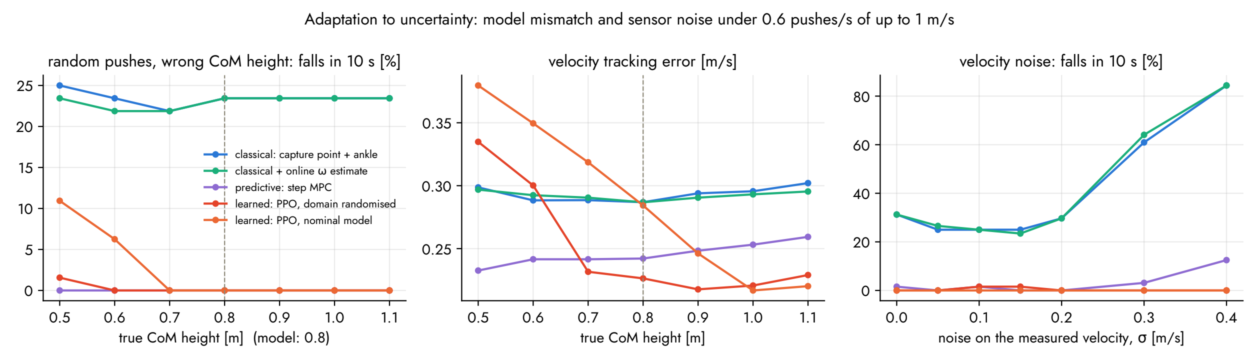

Figure 2: Adaptation to uncertainty. Left: falls under random pushes as the true centre of mass departs from the modelled 0.80 m: at 0.60 m the nominal-only policy falls in 6.25 % of episodes, the randomised one in 0.00 %, the predictive controller in 0.00 %, the classical law in 23.40 % (as at every height). Middle: the velocity error grows for the model-based laws as their periodic offset b becomes wrong. Right: sensor noise; at $\sigma = 0.20$ m/s the classical law falls in 29.70 %, the randomised policy in 0.00 %.

Findings. (1) What the classical law lacks is timing and constraint handling, not the model: estimating ω online changes its curves by nothing measurable, while choosing when to step (the predictive controller) roughly doubles the survivable push. (2) The predictive controller and the learned policy land in the same place by different routes, one reasoning on the step-to-step model at run time, the other having encoded millions of such decisions offline; this is the survey's optimal-

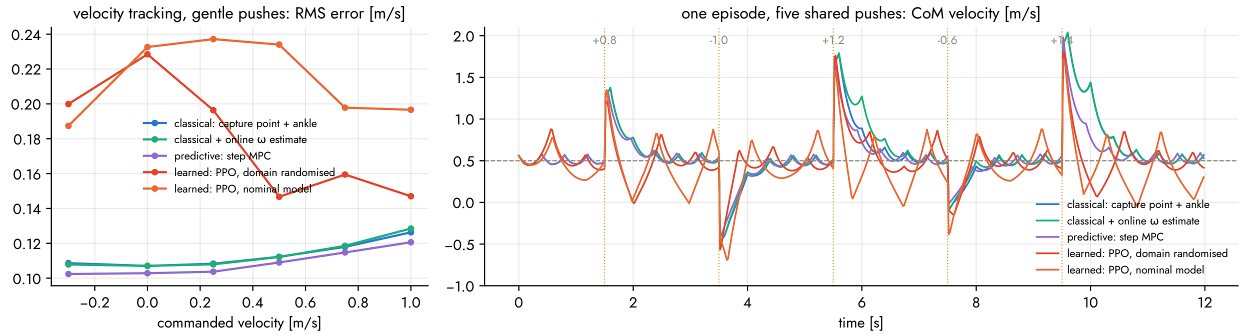

Figure 3: Velocity tracking across commands (left) and the shared five-push episode (right); the classical walker falls at no s, the predictive one at no, the randomised policy at no, the nominal-only policy at no (“no” = stands).

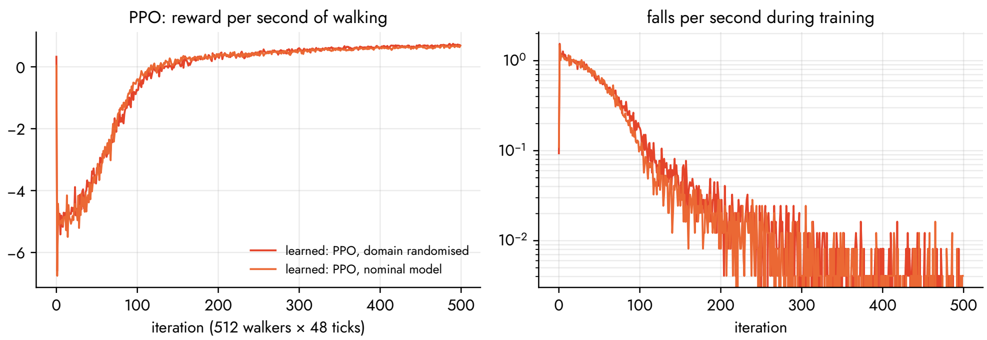

Figure 4: PPO training curves of the two policies.

control reading of RL (its Eqs. 2 and 3). (3) Domain randomisation, the robust-control row of the survey’s Table 2, helps at the edges (97.00 % against 80.00 % survival of 2.20 m/s shoves; 2.00 % against 11.00 % falls on a 0.50 m body), but on this model a well-converged nominal-only policy also stands on the wrong body: with four past velocity samples in its observation it infers the body implicitly, the mechanism the survey attributes to teacher–student and adaptation modules. The first training seed of the nominal policy was a different story (41.00 % falls at 0.60 m); seed variance exceeded the effect of the training distribution, an instance of the algorithmic fragility the survey names. (4) Policies trained with noisy, delayed velocity use their history and do not fall at $\sigma = 0.40\,\mathrm{m/s}$, where the classical law falls 29.70 % of the time. (5) Learning costs tracking accuracy here: 0.15 m/s to 0.23 m/s RMS against 0.11 m/s for the model-based laws.

4 Implementation and Usage

`lipwalk/model.py`: `LIPWorld` (vectorised, exact integration, pushes, noise, delay, randomisation); `controllers.py`: the four controllers and the action decoding shared with training; `sim.py`: `rollout` and `push_recovery`; `train_rl.py`: PPO in PyTorch, exporting the policy to JSON with its observation normalisation; `run_study.py` → `out/study.json`; `make_figures.py` → figures, `assets/data/locomotion_results.json`, `numbers.tex`; `make_videos.py` → the page’s clips. Tests (`pytest -q tests`) check the exact map against the analytic solution, that the periodic gait is a fixed point of the classical law, the MPC’s step limit, and the estimator’s convergence. The browser lab (`assets/js/locomotion_lab.js`) is a line-for-line port running the exported weights at 100 Hz.

5 Limitations

- A planar point-foot LIP with instantaneous, lossless steps and an ankle of 5.00 cm authority: no swing-leg dynamics, torque or velocity limits on the swing, no impacts, no terrain, no lateral dynamics, no perception. The lab shows the mechanism of each paradigm under the same disturbance, not humanoid performance.
- The classical and predictive controllers are one reasonable design each, not the best of their eras; the classical law's fixed period is the choice that makes the comparison legible. The MPC solves a box-constrained least-squares problem, a stand-in for a QP with contact constraints.
- The learned policies are trained on this very model, so "sim-to-real" here is model mismatch inside the same family; real deployment adds everything listed above.
- Survival curves are computed with random step phase at the push and one seed per protocol; differences of a few percent between controllers are within the sampling noise of 64 walkers. Two PPO seeds per policy were trained and the better on the nominal body kept; with more seeds the learned curves would move by several percent.

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